

Crandall's Crib

INTRO TO AI CRASH COURSE

CS 470

2026

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What is AI Discussion

Is there a “correct” approach to building AI of the four camps?

Will we ever have “true AI?”

Does the Chinese Room argument matter?

READING

TIMELINE

01

Discussion

02

Agents

Agents and
Decision Theory

03

Fields Mention

04

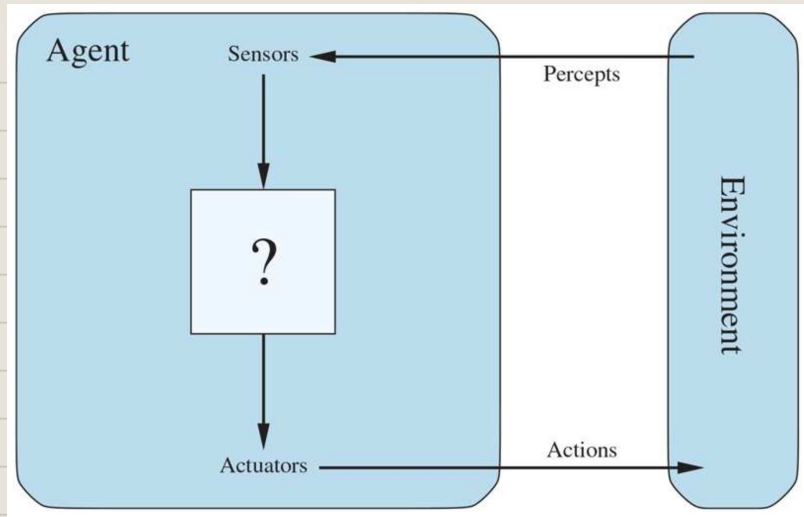
Game Theory



Agents

An agent is defined as:

1. One that acts or exerts power
2. One who makes decisions
3. Perceives environment through sensors
4. Acts through actuators



Types of Agents

- Reflexive
- Goal-based
- Utility-based
- Learning agents
- etc

ENVIRONMENTS

WHERE and WHAT

Observable

Where decisions are made based on a complete knowledge or partial knowledge.

Fully Observable v Partially Observable

Deterministic

A setting where the next state is completely predictable with no randomness or uncertainty.

Deterministic v Non Deterministic

Episodic

The agent needs no prior knowledge or experience to make a choice in a new state.

Episodic v Sequential (Simpsons v Gravity Falls)

Static

The state of the environment is unchanged while the agent deliberates.

Static v Dynamic (Chess v Driving)

Discrete

State and action spaces are a finite set of values. Can use a search tree or decision tree.

Discrete v Continuous (Chess v Driving)

Single-agent

The agent is the only one acting on the environment, or do they have to work around/with other agents.

Single v Multi agent (Solitaire v Catan)

- ☐
- ☐
- ☐
- ☐
- ☐
- ☐

ELEMENTS OF DECISION MAKING

The components that shape making a decision

States

State of the environment the agent operates in. Conditions that exist in the world that determine what happens when an agent acts.

Actions

Set of actions available to the agent, can depend on state.

Consequence

Result of choosing an action in a particular state. When we choose an action, something will happen.

Utility/Cost

Numerical representation for how well a consequence accomplishes a goal. Tied to consequences

Goal

What the agent is trying to achieve. Usually wants to maximize reward/utility

USING UTILITIES TO MAKE DECISIONS

Risk Seeking

LATER TOPIC

Risk Neutral

Choose the best

$$\bar{u}(a) = \sum_s u(s, a)p(s)$$

$$a^* = \arg \max_a \bar{u}(a)$$

RATIONAL AGENT

Risk Averse

Choose the best worst-case

$$a^* = \arg \max_a \min_s u(s, a)$$

SECURITY POLICY

OUTCOMES

01. **RISK NEUTRAL**

Willing to accept some losses if it means you can get higher gains, on average.

In other words, maximize expected utility.

Not perfect, but often what we focus on.

02. **RISK AVERSE**

Afraid to take any risk, assume worst-case scenarios, choose the best one.

03. **RISK SEEKING**

Prefers uncertain gambles with positive variance over a guaranteed outcome with the same expected value. (Convex utility function)

BUILDING A UTILITY FUNCTION

See more in the Utility and Decisions section of the slides.

In essence, determine the value of the states your agent wants to be in and encourage it to choose the best state given the environment.

Work custodial $u=4$ or work grounds $u=8$
Allergic to grass? Hospital $u=-30$
Choose custodial, better outcome

Oversimplified concept

TIMELINE

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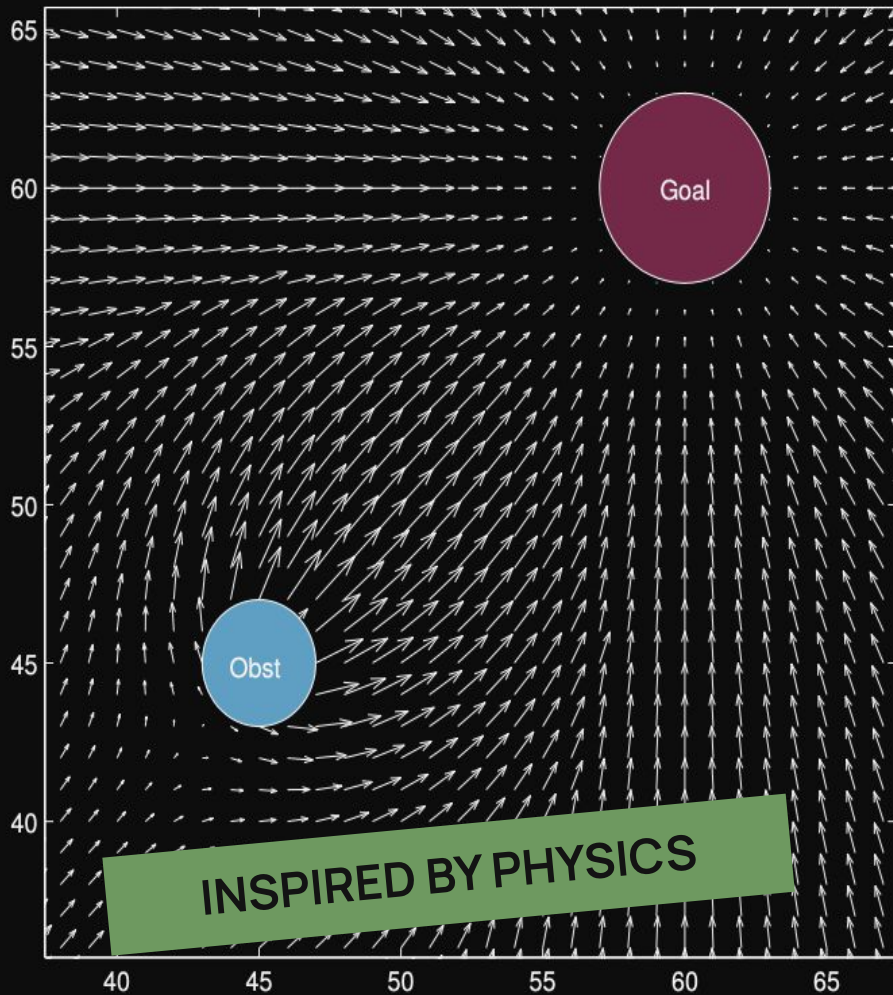
03

Fields Mention

Reflex Agent
Example

04

Game Theory



POTENTIAL FIELDS

Represent a way to guide a robot using attractive and repulsive forces

Minimize a cost function to find an optimal path based on attractive and repulsive forces

example of a reflex agent

REFLEX VS. UTILITY AGENTS

PRE PLANNING

The agent has the map and actions as soon as the fields are available.

Compute is then just about asserting the state and executing the predefined action for that state.

LOCAL MINIMA

Potential fields can have places of local minima depending on the calculation policy, the agent can find itself stuck with no capacity to get unstuck.

STRATEGY

Reflex agents either have attractive states or repulsive states, cannot strategically take a loss for a better end result.

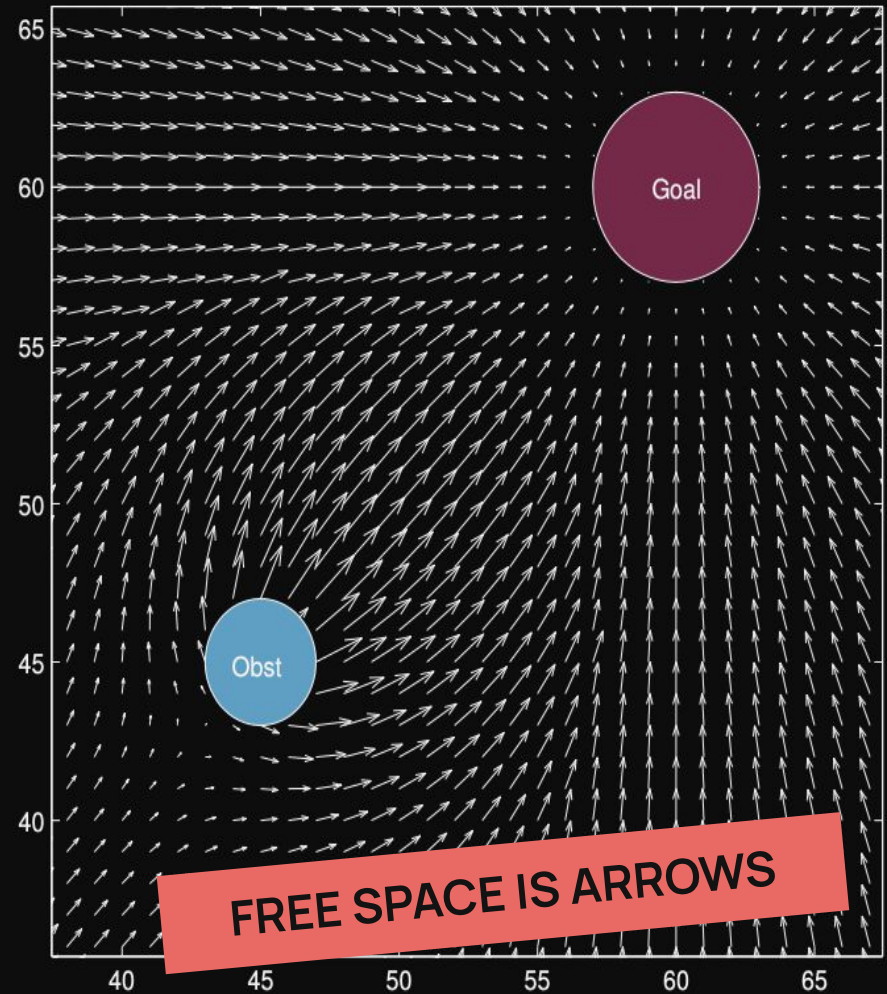
POLICY V OUTCOME

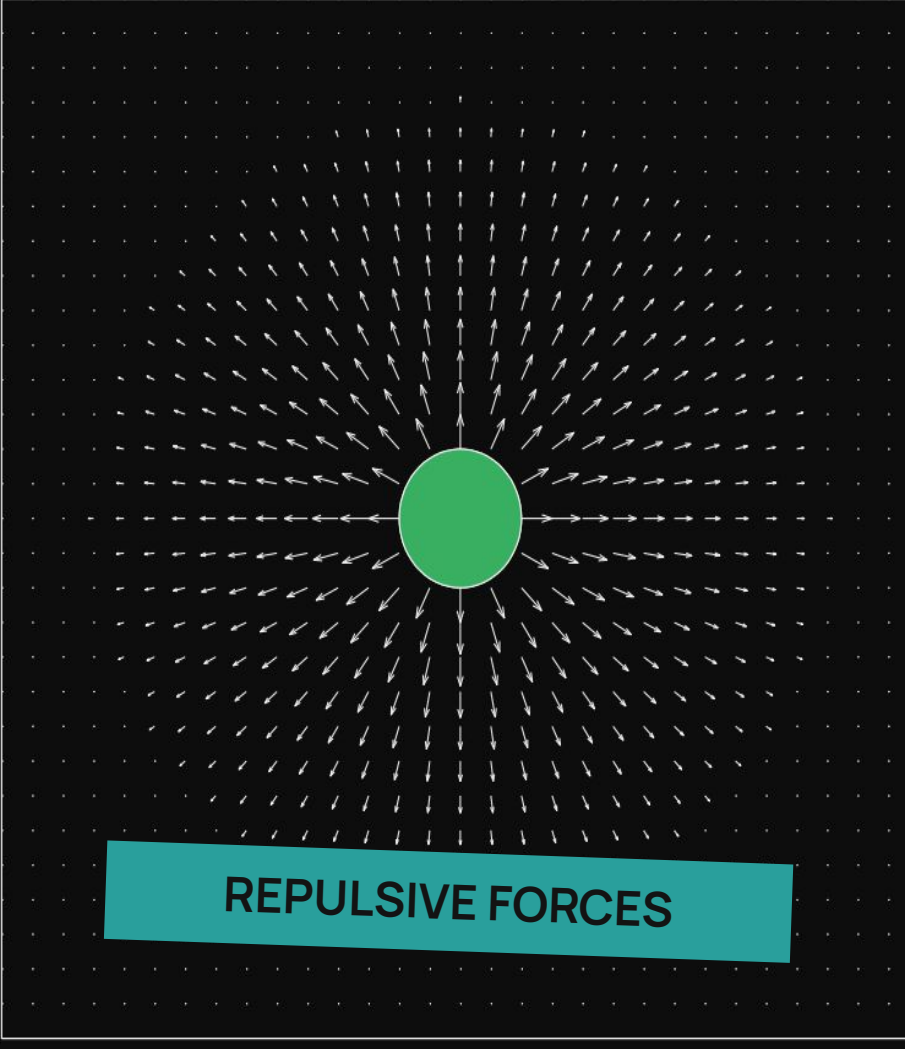
Acts according to an established rule set. Utility agents are trying to get to the state with the best consequences. Reflex agents are working the quota vs applying logic to get to a desired end. Doctor who has to prescribe according to insurance policy v cancer doctor evaluating how to get to desired patient outcome.

CONFIGURATION SPACE

Represents all possible positions and orientations of the robot

Divided into free space and obstacle space



A diagram illustrating repulsive forces. It features a central green circle on a black background with a white grid. Numerous white arrows radiate outwards from the circle, representing the direction and magnitude of the repulsive force. The arrows are longer closer to the circle and shorter further away.

REPULSIVE FORCES

FORCES

Attractive and repulsive

Attractive:

- Goal
- Other elements like recharge stations

Repulsive:

- Obstacles
- Walls
- Holes
- Explosive mines

These forces can have scalar and negative values.

FORCE EQUATIONS

$$\mathbf{v} = [x, y]^T \rightarrow \Delta = [\Delta x, \Delta y]^T$$

DISTANCE

$$d = \sqrt{(x_G - x)^2 + (y - y_G)^2}$$

THETA

$$\theta = \tan^{-1} \left(\frac{y_G - y}{x_G - x} \right)$$

ATTRACTIVE

if $d < r$

$$\Delta x = \Delta y = 0$$

if $r \leq d \leq s + r$

$$\Delta x = \alpha(d - r) \cos(\theta)$$

$$\Delta y = \alpha(d - r) \sin(\theta)$$

if $d > s + r$

$$\Delta x = \alpha s \cos(\theta)$$

$$\Delta y = \alpha s \sin(\theta)$$

REPULSIVE

if $d < r$

$$\Delta x = -\text{sign}(\cos(\theta)) \infty$$

$$\Delta y = -\text{sign}(\sin(\theta)) \infty$$

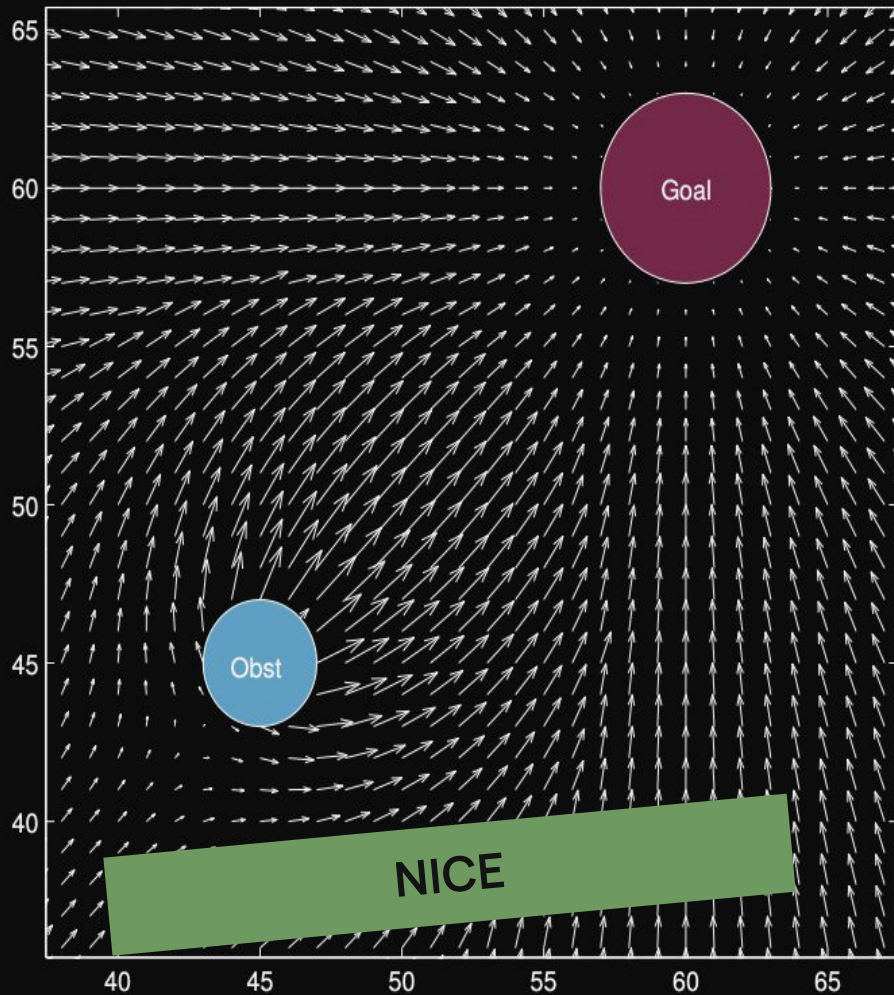
if $r \leq d \leq s + r$

$$\Delta x = -\beta(s + r - d) \cos(\theta)$$

$$\Delta y = -\beta(s + r - d) \sin(\theta)$$

if $d > s + r$

$$\Delta x = \Delta y = 0$$



COMBINE FORCES

Sum the attractive and
repulsive forces

ALGORITHM

Initialize:

- Set initial and goal positions
- Define obstacle positions

Update Potential Fields:

- Calculate attractive and repulsive potentials

Compute Total Force:

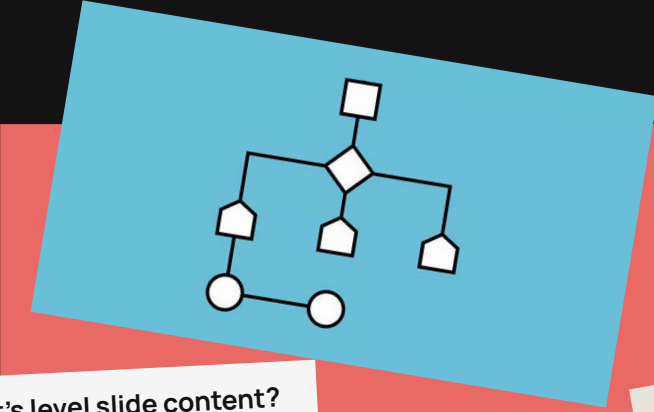
- Sum attractive and repulsive forces

Update Robot's Position:

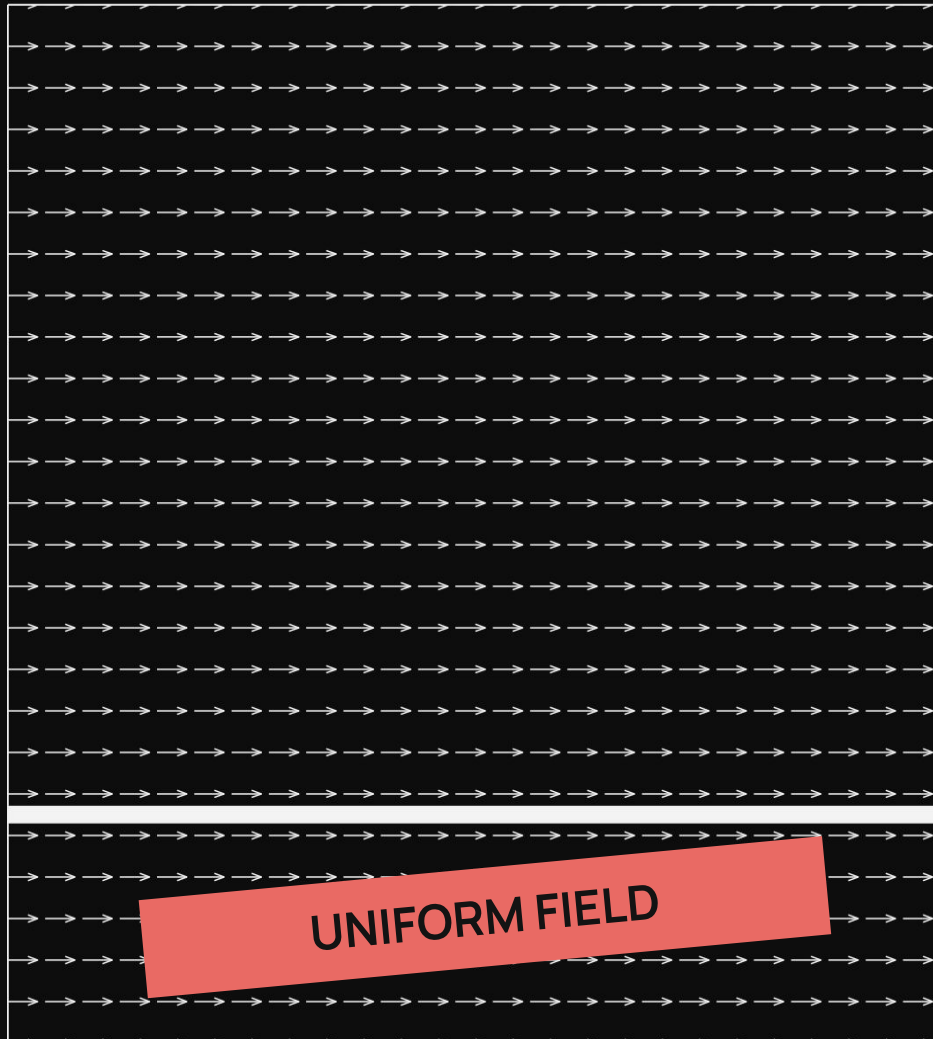
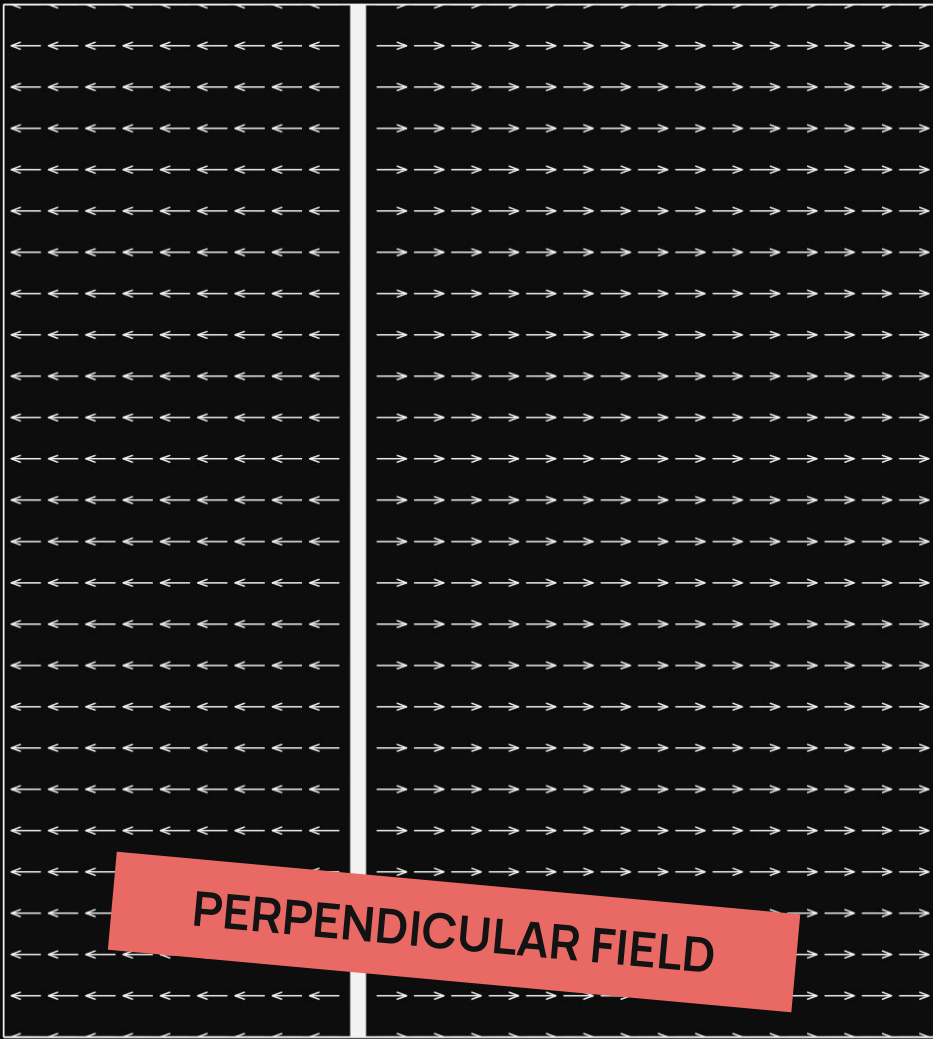
- Move the robot in the direction of the total force

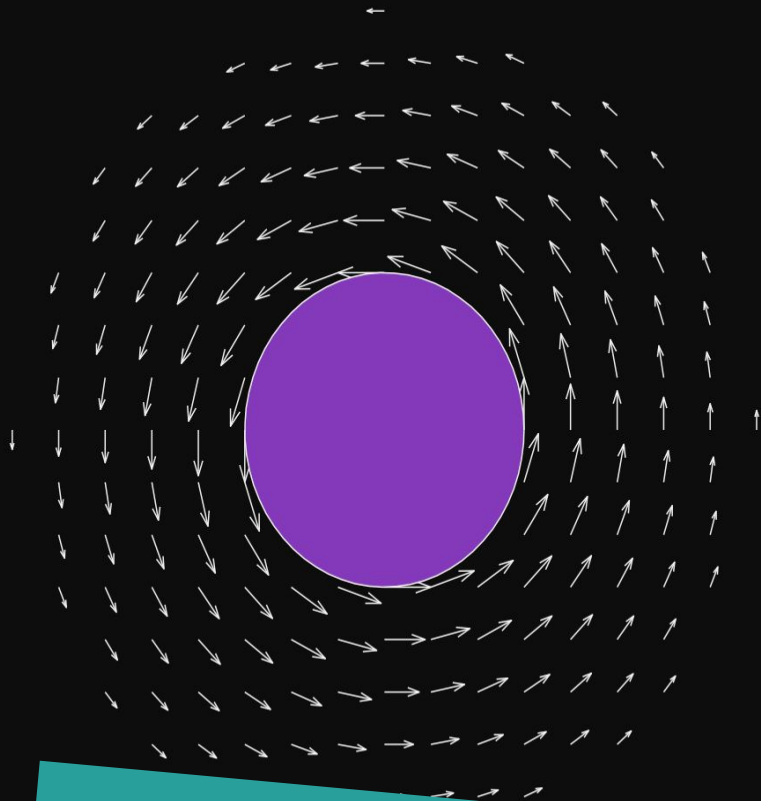
Repeat:

- Iterate until the robot reaches the goal or a predetermined condition is met



what's level slide content?





TANGENTIAL FIELD



RANDOM FIELD

STRENGTHS

Relatively simple

Easy to implement

Very effective in static environments

WEAKNESSES

Repulsive forces might prevent robot from reaching goal

Ex: Goal is in between a narrow alleyway

Might get stuck in local minima

WHEN TO USE REFLEX AGENT

The environment is fully observable

The environment is deterministic

There is little time for deliberation (i.e. we must act quickly/instantaneously)

No learning or adaptation is required

TIMELINE

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Game Theory

It's just a theory...

GAME THEORY

- A way to treat problems as games
- Create mathematical models of interactions between “players”
- Players are usually viewed as rational agents
- What is the definition of a rational agent?
 - Chooses actions to maximize expected utility

Why Game Theory?

- Often it is easier to think of a problem in terms of a game
- Usually more fun to model problems as a game
- Long history of mathematical models and results we can use
- From an engineering perspective, it can provide insight into system design based on how users make decisions

THE GAME

Set of players

- Assumed to be rational agents

Set of actions

- Moves available to agents during the game

Policy

- Strategy

Reward function (or cost function)

In games, we usually like to arrive at a “good” outcome. In multiplayer games, this looks like:

Pareto Optimum

- Pareto dominated: the utility of all players can be increased
- Pareto optimum: not Pareto dominated
- Best we can hope to achieve

Nash Equilibrium

- Best response: given what other players do, a best response maximizes utility for a given player
- Nash equilibrium: when everyone is playing a best response to what everyone else is doing
- Usually not as good as a Pareto optimal outcome (though sometimes they are the same)
- Usually what players converge to

GAME TYPES

SIMULTANEOUS

Players choose actions at the same time.

SEQUENTIAL

Players take turns when choosing actions.

ZERO-SUM

Players have completely opposite goals. This means that their utilities are opposite.

So, if we add up everyone's utility, the sum is usually 0 (or close to 0).
Cooperation is non-existent.

NON-ZERO-SUM

Players might have similar goals. This means that their utilities are probably similar.

So, if we add up everyone's utility, the sum is usually larger than 0.
Cooperation is more common.

LIMIT- ATIONS

RATIONAL?

We assume all players are rational. Not always the case.

ENVIRONMENT?

Better models require information of the whole game and how others think, not always available.

CONSEQUENCE?

A game is just a game with no real outcomes, boredom leads to irrationality. Not always an exact model of reality.

TIMELINE

05

Minmax

Get the bag.

06

Heuristics

07

Narrow AI

08

Probability

